

Numerical simulation of multi-dimensional NMR response in tight sandstone

Jiangfeng Guo, Ranhong Xie, Youlong Zou and Yejiao Ding

State Key Laboratory of Petroleum Resources and Prospecting, China University of Petroleum (Beijing), Beijing 102249, People's Republic of China

E-mail: xieranhong@cup.edu.cn

Received 7 November 2015, revised 29 February 2016

Accepted for publication 11 March 2016

Published 20 April 2016



Abstract

Conventional logging methods have limitations in the evaluation of tight sandstone reservoirs. The multi-dimensional nuclear magnetic resonance (NMR) logging method has the advantage that it can simultaneously measure transverse relaxation time (T_2), longitudinal relaxation time (T_1) and diffusion coefficient (D). In this paper, we simulate NMR measurements of tight sandstone with different wettability and saturations by the random walk method and obtain the magnetization decays of Carr–Purcell–Meiboom–Gill pulse sequences with different wait times (TW) and echo spacings (TE) under a magnetic field gradient, resulting in D - T_2 - T_1 maps by the multiple echo trains joint inversion method. We also study the effects of wettability, saturation, signal-to-noise ratio (SNR) of data and restricted diffusion on the D - T_2 - T_1 maps in tight sandstone. The results show that with decreasing wetting fluid saturation, the surface relaxation rate of the wetting fluid gradually increases and the restricted diffusion phenomenon becomes more and more obvious, which leads to the wetting fluid signal moving along the direction of short relaxation and the direction of the diffusion coefficient decreasing in D - T_2 - T_1 maps. Meanwhile, the non-wetting fluid position in D - T_2 - T_1 maps does not change with saturation variation. With decreasing SNR, the ability to identify water and oil signals based on NMR maps gradually decreases. The wetting fluid D - T_1 and D - T_2 correlations in NMR diffusion-relaxation maps of tight sandstone are obtained through expanding the wetting fluid restricted diffusion models, and are further applied to recognize the wetting fluid in simulated D - T_2 maps and D - T_1 maps.

Keywords: tight sandstone, multi-dimensional NMR, random walk, wettability, restricted diffusion

(Some figures may appear in colour only in the online journal)

1. Introduction

Tight sandstone reservoirs have the characteristics of low porosity, low permeability and complex pore structure, making evaluating petrophysical properties of these reservoirs by conventional logging methods challenging. Therefore it is necessary to develop new well logging techniques such as nuclear magnetic resonance (NMR) logging to overcome this challenge. NMR logging measures the relaxation and diffusion information of the hydrogen in formation pore fluid, which can be used to obtain petrophysical parameters, such as porosity, permeability, pore structure, wettability and various fluids saturations (George *et al* 1998, Dunn *et al* 2002, Chen

et al 2003). However, one dimensional (1D) NMR logging can only make use of the distribution of transverse relaxation time (T_2), when multi-phase fluids (water, oil and gas) coexist in porous media. T_2 distributions of various fluids often overlap one another, making it difficult to identify the fluids from one another (Sun and Dunn 2002, Hurlimann *et al* 2002a, 2002b, Xie *et al* 2005, Tan *et al* 2014). Multi-dimensional NMR describes the spatial distribution of hydrogen protons by using multiple physical parameters, such as transverse relaxation time (T_2), longitudinal relaxation time (T_1) and diffusion coefficient (D), which can overcome the disadvantage of 1D NMR in reservoir fluids identification (Heaton *et al* 2004, 2008, Xie *et al* 2009a, Xie and Xiao 2009).

Complex pore structure, wettability and various fluids saturations of tight sandstone reservoirs are the key factors that affect their NMR response. The random walk method can be used to simulate the procedure of NMR measurements accurately and effectively, as well as to study the influence of individual factors on the NMR response. There has been quite a lot of previous research on simulating 1D and two-dimensional (2D) NMR responses of sandstones and carbonate rocks; for example, Toumelin *et al* (2004, 2007) simulated the magnetization decays of Carr–Purcell–Meiboom–Gill (CPMG) pulse sequences with different wait times (TW) and echo spacings (TE) based on disordered packs of spherical grains, and systematically studied the influence of various factors on fluid identification in D - T_2 maps and D - T_1 maps. Talabi *et al* (2008), and Arns and Melean (2009) simulated the magnetization decays of single phase fluid in carbonate rocks by the random walk method, and studied the characteristics of water and oil in T_2 distributions. Talabi *et al* (2009, 2010) simulated the magnetization decays of single phase and two phase fluids in sandstone by the random walk method. Arns *et al* (2011), and Alghamdi and Arns (2012) simulated the magnetization decays of the pulsed field gradient stimulated echo (PFGSTE) encoding sequence in Bentheimer sandstone and Berea sandstone, and studied the characteristics of water and oil in D - T_2 maps.

However, few studies have been done on NMR response simulation in tight sandstone, especially multi-dimensional NMR response. In this paper, we simulated NMR measurements of tight sandstones by the random walk method based on digital core, and studied multi-dimensional NMR response of tight sandstone saturated with water and oil.

2. NMR theory

The basis of NMR is the interaction between the magnetic field of the nucleus and the external magnetic field. Before NMR measurements, the hydrogen protons of the formation fluid are orientated randomly and arranged arbitrarily; the spin system has no magnetization macroscopically in this case. Once measurement begins, the permanent magnet of the NMR logging tool produces a static magnetic field $\mathbf{B}_0$ which can make the disordered hydrogen protons of the formation fluid magnetized and in order in the nuclear spin space. The magnetization gradually reaches the equilibrium value M_0 , which has the same direction as $\mathbf{B}_0$.

The antenna of NMR logging tools produces an alternating magnetic field $\mathbf{B}_1$, the direction of which is perpendicular to that of $\mathbf{B}_0$. When the frequency (ω) of the alternating magnetic field is equal to the Larmor frequency (ω_0) of protons, the resonance phenomenon will occur in the nuclear spin system; that is, the nuclear magnetic moment in the low energy state will jump to a high energy state by absorbing energy provided by the alternating magnetic field. Macroscopically, resonance leads to M_0 deviating from $\mathbf{B}_0$, and M_0 rotates 90° to the xy -plane by adjusting the pulse width of the alternating magnetic field. After the alternating magnetic field is removed, only the external magnetic field $\mathbf{B}_0$ in the z -axis direction persists, and M_0 precesses along $\mathbf{B}_0$. The nuclear spin system will

recover from the non-equilibrium state to equilibrium state. Magnetization M_0 is decomposed into two components on the xy -plane (transverse component M_x and M_y) and one component in the z -axis (longitudinal component M_z). The process of xy -plane components decay back to the initial zero state is defined as the transverse relaxation process, and the time consumed is expressed as the transverse relaxation time T_2 . The process of z -axis component recovery back to the initial magnetization M_0 is defined as the longitudinal relaxation process, and the time consumed is expressed as the longitudinal relaxation time T_1 . Bloch (1946) described the changing of M_x , M_y , M_z with time in the rotating frame by

$$\begin{aligned}\frac{dM_x}{dt} &= -\frac{M_x}{T_2}, \\ \frac{dM_y}{dt} &= -\frac{M_y}{T_2}, \\ \frac{dM_z}{dt} &= \frac{M_0 - M_z}{T_1}.\end{aligned}\quad (1)$$

We combine with initial conditions to solve equation (1):

$$\begin{aligned}M_x &= M_y = M_0 \exp(-t/T_2), \\ M_z &= M_0[1 - \exp(-t/T_1)].\end{aligned}\quad (2)$$

According to equation (2), M_x and M_y decay exponentially with a rate of $1/T_2$, while M_z recovers exponentially back to the initial magnetization M_0 with a rate of $1/T_1$.

Longitudinal relaxation of the fluid in porous media is affected by two mechanisms, bulk relaxation and surface relaxation, but transverse relaxation is affected by three mechanisms—bulk relaxation, surface relaxation and diffusion relaxation. T_1 and T_2 of pore fluid are

$$\frac{1}{T_1} = \frac{1}{T_{1B}} + \rho_1 \frac{S}{V}; \quad (3)$$

$$\frac{1}{T_2} = \frac{1}{T_{2B}} + \rho_2 \frac{S}{V} + \frac{(\gamma GE)^2 D}{12}, \quad (4)$$

where T_{1B} and T_{2B} are longitudinal and transverse bulk relaxation times of the fluid, respectively; different fluids have different bulk relaxation times. ρ_1 and ρ_2 are longitudinal and transverse surface relaxivities of rocks, respectively. S/V is the ratio of the surface which relaxes over the volume occupied by that fluid phase. γ is the proton gyromagnetic ratio. G is externally applied magnetic field gradient. TE is the echo spacing. D is the fluid diffusion coefficient and is time dependent in porous media.

3. Random walk method

3.1. Theory of method

The relationship between the macroscopic magnetization of fluid in porous media and time is expressed by Bloch–Torrey equations (Torrey 1956):

$$\frac{d\mathbf{M}}{dt} = D \nabla^2 \mathbf{M} + \gamma \mathbf{B} \times \mathbf{M} - \frac{M_x}{T_2} \mathbf{x} - \frac{M_y}{T_2} \mathbf{y} - \frac{M_z - M_0}{T_1} \mathbf{z}. \quad (5)$$

A CPMG pulse sequence takes the form of a 90° pulse lasting t_{90} along the x -axis followed by a series of 180° pulses lasting t_{180} along the y -axis separated by the echo spacing TE , and spin echoes are generated between two successive 180° pulses. The two major control parameters of CPMG pulse sequence measurements are TW and TE . TW mainly affects the recovery of spin magnetization; if TW is not long enough, some spins may not fully recover to their thermal equilibrium state, i.e. the total magnetization would decrease. TE mainly affects fluid diffusion relaxation and thus the whole magnetization decays. In the process of polarization, the recovery of normalized magnetization with time is calculated using the following equation:

$$\begin{aligned} M_1(t) &= 1 - \exp\left(-\frac{t}{T_{1B}}\right) \cdot \exp\left(-\frac{t}{T_{1S}}\right) \\ &= 1 - \exp\left(-\frac{t}{T_{1B}}\right) \cdot M_{1S}(t). \end{aligned} \quad (6)$$

In the process of decay, the decay of normalized magnetization with time is calculated using the following equation:

$$\begin{aligned} \frac{M_2(t)}{M'_0} &= \exp\left(-\frac{t}{T_{2B}}\right) \cdot \exp\left(-\frac{t}{T_{2S}}\right) \cdot \exp\left(-\frac{\gamma^2 G^2 TE^2 Dt}{12}\right) \\ &= \exp\left(-\frac{t}{T_{2B}}\right) \cdot M_{2S}(t) \cdot M_{2D}(t), \end{aligned} \quad (7)$$

where M'_0 is the initial magnetization when the decay begins and then comes to 1 when the polarization recovers completely.

As for bulk relaxation, surface relaxation and the diffusion relaxation of the fluid in the porous media happens simultaneously and independently (Kenyon 1997). Simulating the decays of NMR magnetization can be achieved through respectively simulating the decays of bulk relaxation magnetization, surface relaxation magnetization and diffusion relaxation magnetization. According to the properties of acquiring CPMG spin echoes, the simulation procedure can be divided into two parts, the first part refers to the simulation of the recovery of the longitudinal component M_z on the z -axis back to its initial magnetization M_0 during the procedure of polarization; the second part refers to the simulation of the decay of the magnetization on the xy -plane, and its initial magnetization equals the recovered magnetization from the first part. The recovery of the longitudinal component M_z on the z -axis back to its initial magnetization M_0 can be regarded as the inverse process of the magnetization decaying along the z -axis. The decay of the magnetization along the z -axis can be simulated by the random walk method. Specific procedures are shown as follows:

- (1) A fixed number of protons are placed randomly in the pore of rocks.
- (2) Calculate the distance (d) between protons and the nearest solid surface, if d is smaller than 3ε (ε is diffusion radius of the conventional random walk method), we choose the conventional random walk method, i.e. diffusion radius $r = \varepsilon$; if d is equal to or larger than 3ε , we choose the

first-passage-time technique (Zheng and Chiew 1989), i.e. diffusion radius $r = d$.

- (3) Calculate the time interval $\Delta t = r^2/6D$ and the position of the proton in the next moment $[x(t + \Delta t), y(t + \Delta t), z(t + \Delta t)]$:

$$\begin{aligned} x(t + \Delta t) &= x(t) + r \cdot \sin \theta \cdot \cos \varphi, \\ y(t + \Delta t) &= y(t) + r \cdot \sin \theta \cdot \sin \varphi, \\ z(t + \Delta t) &= z(t) + r \cdot \cos \theta, \end{aligned} \quad (8)$$

where θ should be selected randomly in the range $[0, \pi]$, φ should be selected randomly in the range $[0, 2\pi]$, $[x(t), y(t), z(t)]$ is the position of the proton in the present moment.

- (4) Check whether collision happens between protons and the solid surface. If it does happen, protons will be killed with a probability given by (Bergman et al 1995):

$$\delta_1 = \frac{2}{3} \frac{\rho_1 r}{D} \times 0.96. \quad (9)$$

If the proton survives, it bounces off the solid surface and returns to its previous position. In the process of simulation, surface relaxation magnetization $M_{1S}(t)$ decays along the z -axis direction:

$$M_{1S}(t) = \frac{N_1(t)}{N_1(0)}, \quad (10)$$

where $N_1(t)$ represents the number of surviving protons at time t and $N_1(0)$ represents the number of initial protons.

- (5) Check whether or not the protons run out of rocks. If this happens, they will land in a random position within the opposite side of rocks.
- (6) Repeat steps (2)–(5) until the sampling time is larger than TW , and record $M_{1S}(TW)$ at that time. Finally, we get the initial magnetization:

$$M'_0 = 1 - \exp\left(-\frac{TW}{T_{1B}}\right) \cdot \frac{N_1(TW)}{N_1(0)}. \quad (11)$$

The simulation of the magnetization decay on the xy -plane is similar to that on the z -axis, only with one added simulation of diffusion relaxation. The specific steps are as follows:

Steps (1)–(5) are similar to those steps mentioned above, only in step (5) the killing probability is changed to:

$$\delta_2 = \frac{2}{3} \frac{\rho_2 r}{D} \times 0.96. \quad (12)$$

- (7) The proton dephasing in Δt time can be calculated by:

$$\begin{aligned} \phi(t + \Delta t) &= \phi(t) + \gamma G \left\{ \frac{z(t + \Delta t) - z(t) - 2z(0)}{2} \right\} \\ &\quad \times \Delta t + \gamma G \sqrt{\frac{D \Delta t^3}{6}} \text{Normal}(), \end{aligned} \quad (13)$$

Table 1. Simulated parameters in the spherical pore.

Radius	Bulk relaxation time ($T_{1B} = T_{2B}$)	Static magnetic field gradient G	Echo spacing TE	Bulk diffusivity D	Longitudinal surface relaxivity ρ_1	Transverse surface relaxivity ρ_2
1 μm	3.0 s	0.3 T m^{-1}	1.2 ms	$2.0 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$	20 $\mu\text{m s}^{-1}$	30 $\mu\text{m s}^{-1}$

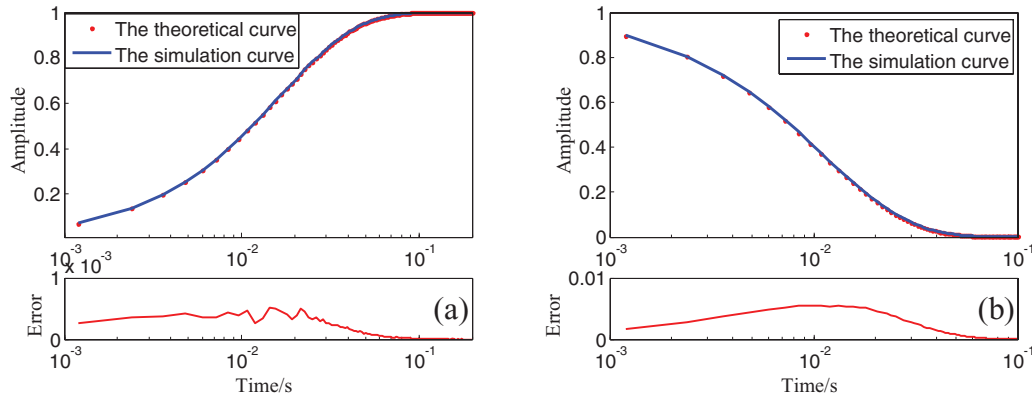


Figure 1. Comparison of the theoretical curve with the simulation curve. (a) Comparison of the magnetization recovery curve simulated from the random walk method with the theoretical curve calculated from equation (16). (b) Comparison of the magnetization decay curve simulated from the random walk method with the theoretical curve calculated from equation (17).

where $\text{Normal}()$ is a normal Gaussian function. In order to conform to the requirements of CPMG pulse sequence acquisition, a phase-shift reversal is passed during the walk step, when $t = (n + 1/2)TE$, and the echo signal is acquired when $t = nTE$. Consequently, the diffusion relaxation magnetization is computed by taking the cosine of all spins' phase shifts, defined by:

$$M_G(nTE) = \frac{1}{N} \sum_{i=1}^N \cos[\phi_i(nTE)]. \quad (14)$$

- (8) Repeat steps (2)–(6) until the sampling time is larger than the set threshold, and record magnetization at nTE time. The normalized magnetization is described by:

$$M_2(nTE) = M'_0 \exp\left(-\frac{nTE}{T_{2B}}\right) \frac{N_2(nTE)}{N_2(0)} \frac{1}{N} \sum_{i=1}^N \cos[\phi_i(nTE)], \quad (15)$$

where $N_2(t)$ represents the number of surviving protons at time t and $N_2(0)$ represents the number of initial protons during the magnetization decay on the xy -plane.

3.2. Verification of method

The accuracy of the random walk method is verified by comparing the results of theoretical calculation and numerical simulation in the spherical pore. The recovery of normalized magnetization is calculated by the theoretical equation:

$$M_1(t) = 1 - \exp\left[-t\left(\frac{1}{T_{1B}} + \rho_1 \frac{S}{V}\right)\right]. \quad (16)$$

After complete polarization, the decay of normalized magnetization with time is also calculated by the theoretical equation:

$$M_2(t) = \exp\left\{-t\left[\frac{1}{T_{2B}} + \rho_2 \frac{S}{V} + \frac{1}{12}(\gamma \cdot G \cdot TE)^2 \cdot D\right]\right\}. \quad (17)$$

Simulation parameters in the spherical pore are shown in table 1. The magnetization recovery curve simulated from the random walk method is compared with the theoretical curve calculated from equation (16), shown in figure 1(a). The magnetization decay curve simulated from the random walk method is compared with the theoretical curve calculated from equation (17), shown in figure 1(b). We evaluated the accuracy of the numerical simulation method based on the difference between the theoretical value and simulation value of magnetization, and the difference is calculated by:

$$\text{Error} = M_S - M_T, \quad (18)$$

where M_S and M_T represent the simulation value and theoretical value of magnetization, respectively. According to figure 1, we can see that the difference is less than 0.01, which suggests the random walk method can be applied to accurately simulate NMR measurements.

4. NMR response of tight sandstone

4.1. Models of the digital core

The digital core technology has been widely used in studying petrophysical numerical simulations in porous media in which constructing an accurate digital core is its basis and key. At present, the method for constructing a digital core mainly includes the numerical reconstruction method and physical experimental method (Arns 2002, Jin et al 2003). The numerical reconstruction method reconstructs the 3D digital core mainly based on the statistical function of 2D thin sections, but cannot guarantee the connectivity of the digital core.

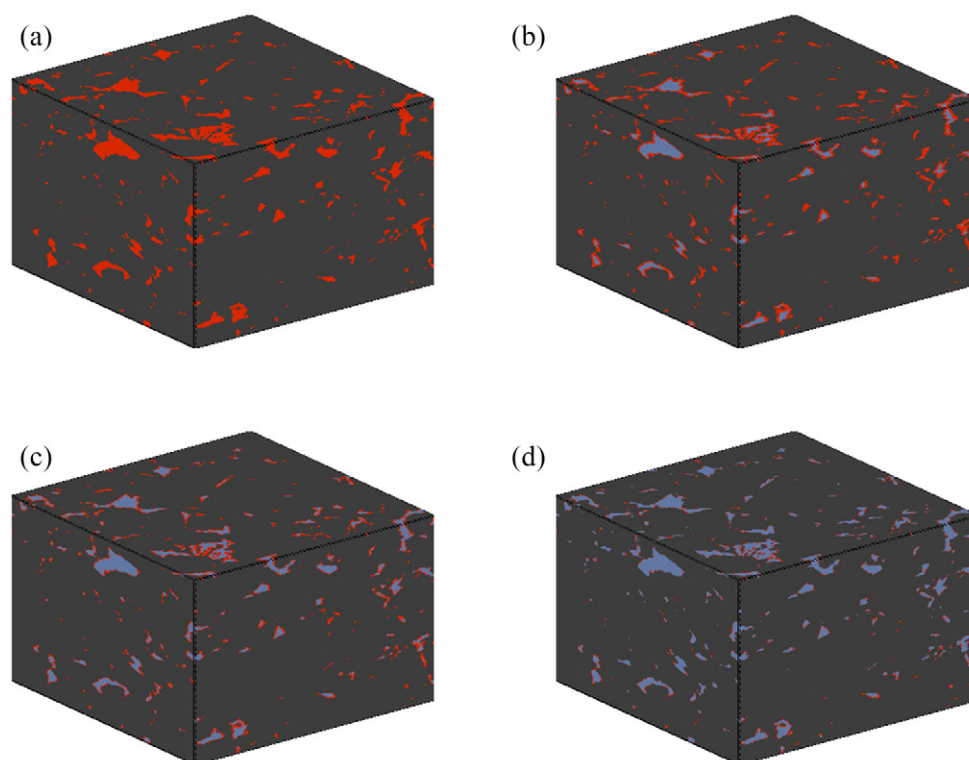


Figure 2. 3D images of tight sandstone with different fluid saturations. (a) An image of tight sandstone fully saturated with wetting fluid. (b) An image of tight sandstone saturated with 75% wetting fluid and 25% non-wetting fluid. (c) An image of tight sandstone saturated with 50% wetting fluid and 50% non-wetting fluid. (d) An image of tight sandstone saturated with 25% wetting fluid and 75% non-wetting fluid.

The x-ray-CT scanning method is the most popular physical experimental method to construct the digital core, which is the most direct and accurate method and has an advantage of measuring the rock samples nondestructively.

The digital core models of tight sandstone used in this paper were obtained by the x-ray-CT scanning method, as shown in figure 2(a). Its voxel size is 600^3 , resolution is $2.77 \mu\text{m}$ and porosity is 6.26%. Imaging models of tight sandstone with different fluid saturations are shown in figures 2(b)–(d), in which the black area represents the rock matrix, red area represents the wetting fluid and blue area represents the non-wetting fluid in the porous media.

4.2. NMR response of tight sandstone in the water-wet case

Water wettability is assumed preserved in tight sandstone, where longitudinal surface relaxivity (ρ_1) is set to $20 \mu\text{m s}^{-1}$ and transverse surface relaxivity (ρ_2) is set to $30 \mu\text{m s}^{-1}$. Saturating fluid properties of tight sandstone are summarized in table 2. The magnetization decays of CPMG pulse sequences with different parameters are simulated using the random walk method under a static magnetic field gradient of 0.3 T m^{-1} , and the parameters are shown in table 3.

The Gaussian noise of the SNR is 100 and is added to obtain noisy echo trains data, and D - T_2 - T_1 maps of tight sandstone with different water saturations in the water-wet case are obtained by the multiple echo trains jointing inversion method (Xie *et al* 2009b). In order to visualize the D - T_2 - T_1 3D maps, they are projected onto D - T_2 , D - T_1 and T_1 - T_2 2D maps respectively, as shown in figure 3. The blue horizontal line

Table 2. Saturating fluid properties.

Fluid	Hydrogen index	Bulk relaxation time ($T_{1B} = T_{2B}$)	Bulk diffusivity D
Water	1.0	3.0 s	$2.0 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$
Oil	1.0	1.0 s	$5.0 \times 10^{-10} \text{ m}^2 \text{ s}^{-1}$

Table 3. CPMG pulse sequence parameters.

No.	Wait time TW (s)	Echo spacing TE (ms)	Echo number NE
1	12.0	0.45	4000
2	3.0	0.45	4000
3	1.0	0.45	4000
4	0.3	0.45	4000
5	0.1	0.45	4000
6	0.03	0.45	4000
7	0.01	0.45	4000
8	0.003	0.45	4000
9	1.0	0.9	2000
10	1.0	1.8	1000
11	1.0	3.6	500
12	1.0	7.2	250
13	1.0	9.6	187
14	1.0	12.5	144

represents the water line and the green inclined line represents the oil line in D - T_2 and D - T_1 maps, with the blue diagonal line representing $T_1 = T_2$ in T_1 - T_2 maps.

Figure 3 shows that the intensity of the water signal gradually decreases while the opposite behavior is demonstrated for

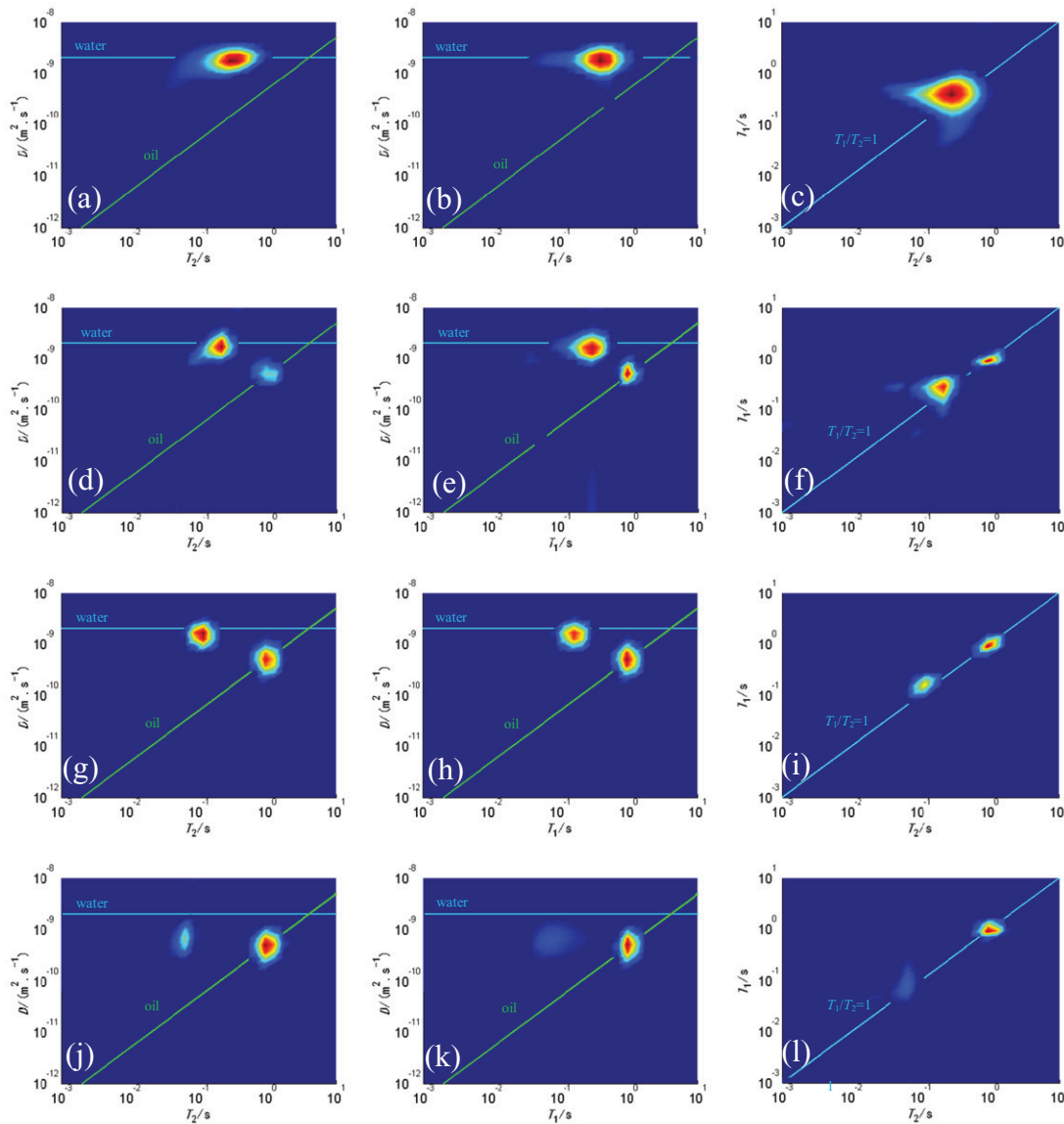


Figure 3. 3D NMR response of tight sandstone with different water saturation in the water-wet case. (a)–(c) D - T_2 map, D - T_1 map and T_1 - T_2 map of tight sandstone fully saturated with wetting water. (d)–(f) D - T_2 map, D - T_1 map and T_1 - T_2 map of tight sandstone saturated with 75% wetting water and 25% non-wetting oil. (g)–(i) D - T_2 map, D - T_1 map and T_1 - T_2 map of tight sandstone saturated with 50% wetting water and 50% non-wetting oil. (j)–(l) D - T_2 map, D - T_1 map and T_1 - T_2 map of tight sandstone saturated with 25% wetting water and 75% non-wetting oil.

the intensity of the oil signal, with water saturation decreasing and oil saturation increasing. The position of the oil signal in D - T_2 - T_1 maps does not change with changing oil saturation. However, with water saturation decreasing, the water signal gradually moves away from the water line and shifts to the short relaxation time in D - T_2 and D - T_1 maps, gradually moving to the short relaxation time along the diagonal direction in T_1 - T_2 maps. This is because the ratio of the effective surface area and volume of pore space in the tight sandstone occupied by the wetting water gradually increases with decreasing water saturation, i.e. the surface relaxation rate of wetting water gradually increases, and as a result, the wetting water signal moves to the short relaxation time in D - T_2 - T_1 maps. In addition, the restricted diffusion phenomenon of wetting water becomes more and more obvious with decreasing water saturation, resulting in the wetting water signal moving

away from the water line to the direction of decreasing diffusion coefficient in D - T_2 - T_1 maps.

4.3. NMR response of tight sandstone in the oil-wet case

Oil wettability is assumed preserved in tight sandstone, with longitudinal surface relaxivity (ρ_1) set to $7 \mu\text{m s}^{-1}$ and transverse surface relaxivity (ρ_2) set to $10 \mu\text{m s}^{-1}$. Saturating fluid properties of tight sandstone are summarized in table 2. The magnetization decays of CPMG pulse sequences with different parameters are simulated using the random walk method under a static magnetic field gradient of 0.3 T m^{-1} , and the parameters are shown in table 3. Gaussian noise of SNR is 100 and is added to obtain noisy echo trains data. D - T_2 - T_1 maps of tight sandstone with different oil saturation in the oil-wet case are obtained by the multiple echo trains jointing inversion

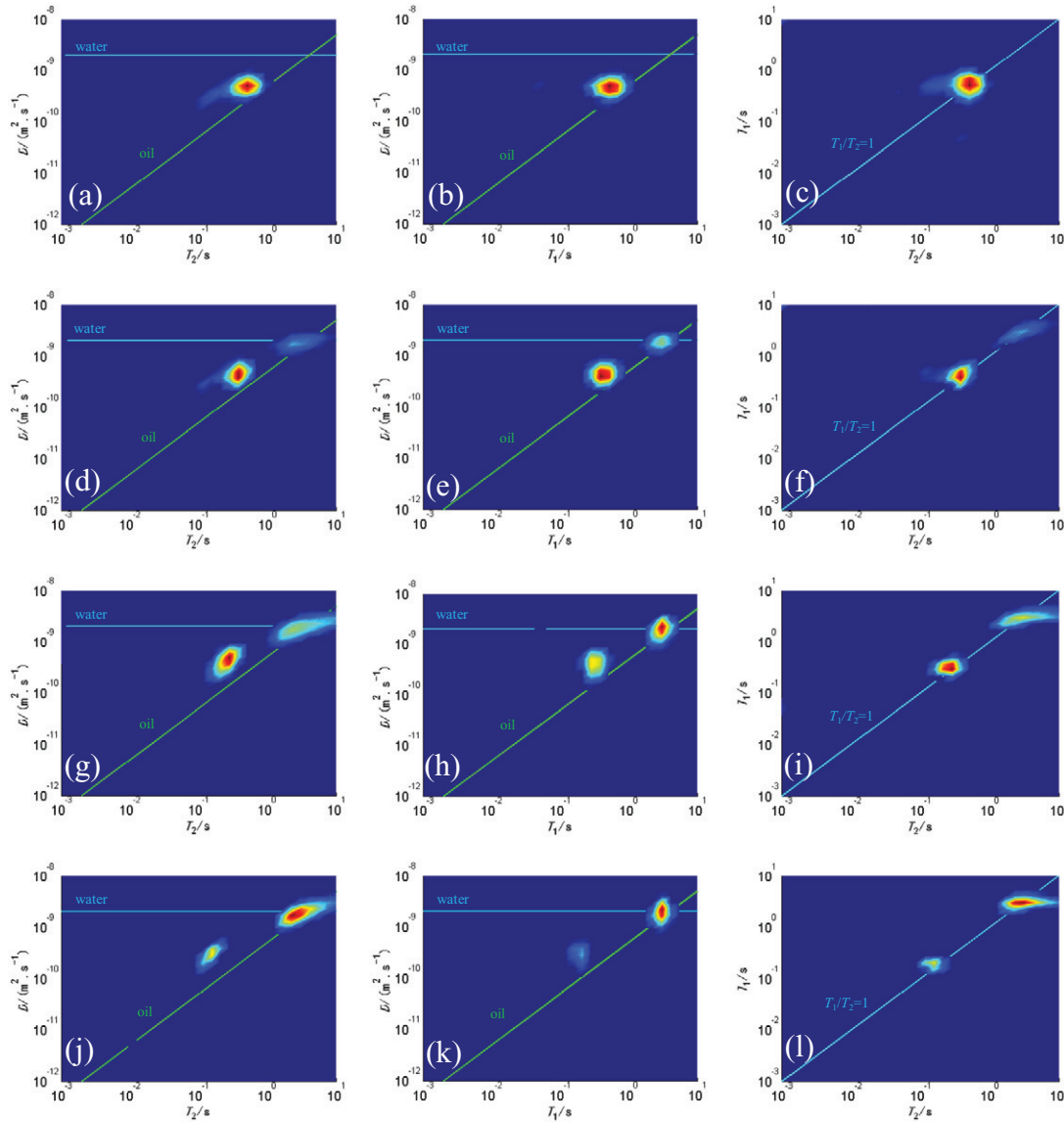


Figure 4. 3D NMR response of tight sandstone with different oil saturation in the oil-wet case. (a)–(c) D - T_2 map, D - T_1 map and T_1 - T_2 map of tight sandstone fully saturated with wetting oil. (d)–(f) D - T_2 map, D - T_1 map and T_1 - T_2 map of tight sandstone saturated with 75% wetting oil and 25% non-wetting water. (g)–(i) D - T_2 map, D - T_1 map and T_1 - T_2 map of tight sandstone saturated with 50% wetting oil and 50% non-wetting water. (j)–(l) D - T_2 map, D - T_1 map and T_1 - T_2 map of tight sandstone saturated with 25% wetting oil and 75% non-wetting water.

method. The D - T_2 - T_1 3D maps are projected onto D - T_2 , D - T_1 and T_1 - T_2 2D maps, respectively, as shown in figure 4. The meanings of various lines in figure 4 are the same as those in figure 3.

Figure 4 shows that the intensity of the oil signal gradually decreases and the intensity of the water signal gradually increases, with oil saturation decreasing and water saturation increasing. The position of the water signal in D - T_2 - T_1 maps does not change with changing water saturation. However, with oil saturation decreasing, the oil signal gradually moves away from the oil line and shifts to the short relaxation time in D - T_2 and D - T_1 maps, and gradually moves to the short relaxation time along the diagonal direction in T_1 - T_2 maps. This is because the ratio of the effective surface area and volume of pore space in the tight sandstone occupied by wetting oil gradually increases with decreasing oil saturation, i.e. the surface

relaxation rate of wetting oil gradually increases, and as a result, the wetting oil signal moves to the short relaxation time in D - T_2 - T_1 maps. In addition, the restricted diffusion phenomenon of wetting oil becomes more and more obvious with decreasing oil saturation, resulting in the wetting oil signal moving away from the oil line to the direction of decreasing diffusion coefficient in D - T_2 - T_1 maps.

Comparing D - T_2 - T_1 maps in figure 3 with those in figure 4, we can see that, in the water-wet case, the oil signal can be identified according to the oil line in D - T_2 maps and D - T_1 maps, while in the oil-wetting case, the wetting oil signal moves away from the oil line in D - T_2 maps and D - T_1 maps. The total relaxation time of the wetting oil is obviously less than its bulk relaxation time because of the influence of surface relaxation. The T_1/T_2 value of water and oil are both close to 1, so the positions of the water and oil signal stand near the

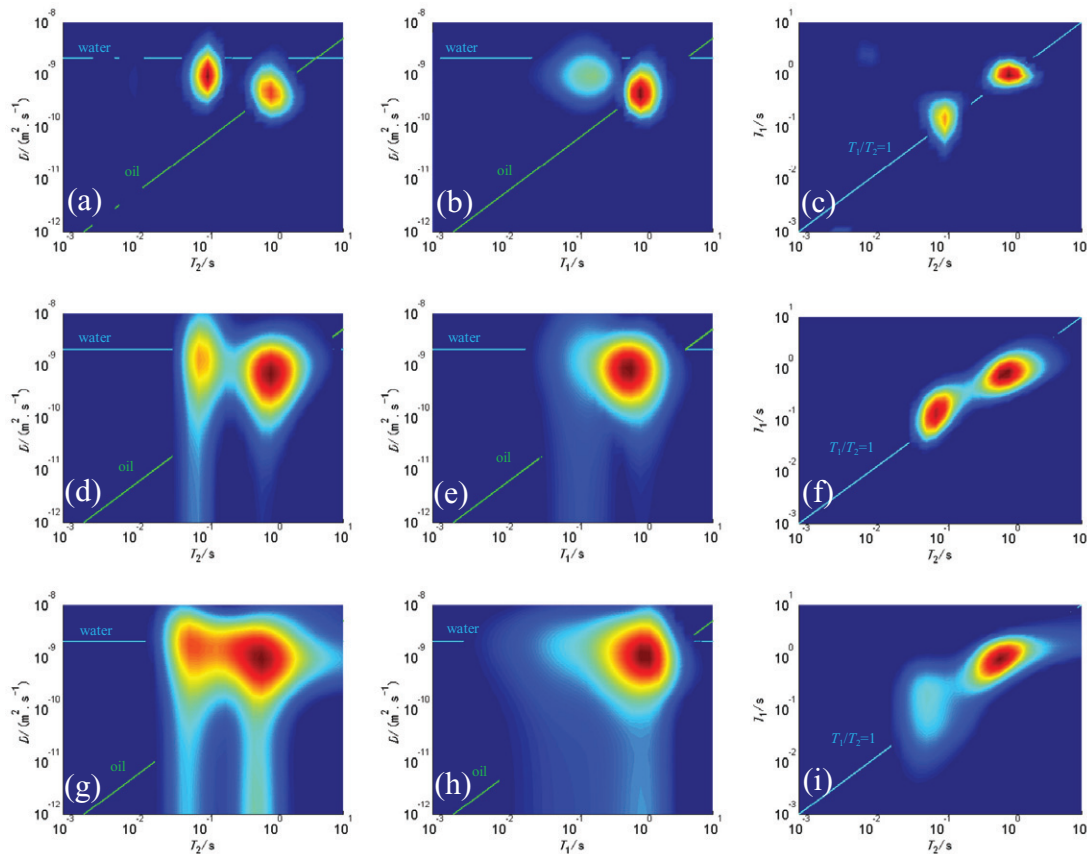


Figure 5. 3D NMR maps of tight sandstone sensitivity to SNR. (a)–(c) D - T_2 map, D - T_1 map and T_1 - T_2 map of tight sandstone saturated with 50% wetting water and 50% non-wetting oil in the case of SNR = 50. (d)–(f) D - T_2 map, D - T_1 map and T_1 - T_2 map of tight sandstone saturated with 50% wetting water and 50% non-wetting oil in the case of SNR = 20. (g)–(i) D - T_2 map, D - T_1 map and T_1 - T_2 map of tight sandstone saturated with 50% wetting water and 50% non-wetting oil in the case of SNR = 10.

diagonal ($T_1/T_2 = 1$) in T_1 - T_2 maps, and their relative positions change with wettability. According to the tight sandstone simulated in this paper and its saturated fluid properties, when the rock is water-wet, the water signal sits to the lower left of the oil signal in T_1 - T_2 maps. In contrast, when the rock is oil-wet, the water signal sits to the top right of the oil signal in T_1 - T_2 maps. So it is not easy to identify water and oil in T_1 - T_2 maps.

4.4. NMR maps of tight sandstone sensitivity to SNR

The fluid signal in tight sandstone detected by NMR logging is very weak, because of low porosity. The SNR of downhole NMR data is low in tight sandstone reservoirs, and data SNR strongly affects the result of NMR data inversion. The echo trains of tight sandstone saturated with 50% wetting water and 50% non-wetting oil obtained in section 4.2 are used as an example. Gaussian noise of different SNR (SNR = 50, 20, 10) is added to echo trains data, and D - T_2 - T_1 maps of tight sandstone with different SNR are obtained by the multiple echo trains jointing inversion method. The D - T_2 - T_1 3D maps are projected onto D - T_2 , D - T_1 and T_1 - T_2 2D maps, respectively, as shown in figure 5. The meanings of various lines in figure 5 are the same as those in figure 3.

As seen in figure 5, we can identify water and oil signals clearly in D - T_2 , D - T_1 and T_1 - T_2 maps, when SNR is 50.

With decreasing SNR, the signals of water and oil gradually overlap one another and appear ‘long tail’ along the direction of decreasing diffusion coefficient in D - T_2 and D - T_1 maps, so the ability of identifying water and oil signals based on NMR maps gradually decreases at this time. The effects of SNR on T_1 - T_2 maps is a little weaker than the effects of SNR on D - T_2 and D - T_1 maps.

5. Restricted diffusion model

In porous media, fluid molecular motion is affected by the presence of pore walls, and diffusion is restricted such that the apparent diffusion coefficient is time dependent and smaller than that of the bulk fluid (Mitra et al 1992, Latour et al 1993, Hurlimann et al 2002a, Luo et al 2015). According to the D - T_2 - T_1 maps of tight sandstone in section 4, restricted diffusion has a great influence on the NMR response of wetting fluid. For the water-wet case, the water signal moves away from the water line, and for the oil-wet case, the oil signal moves away from the oil line in 2D diffusion-relaxation maps, which makes it hard to identify water and oil. So water could be mistaken for oil in the water-wet case and oil could be mistaken for water or oil of higher viscosity in the oil-wet case. We should consider the effects of restricted diffusion when interpreting D - T_2 - T_1 maps of tight sandstone, especially in the small wetting fluid saturation case. The unrestricted diffusion

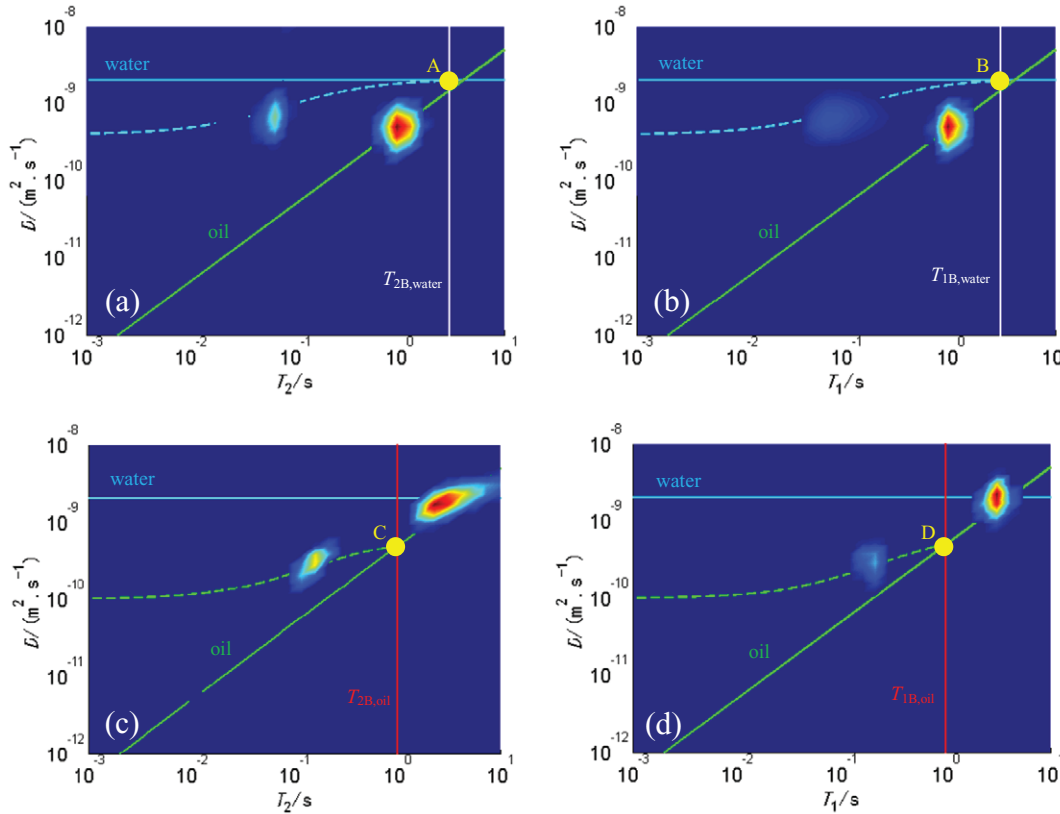


Figure 6. Restricted diffusion model of wetting fluid. (a)–(b) Interpretation for the restricted diffusion water signal in D - T_2 map and D - T_1 map of tight sandstone saturated with 25% wetting water and 75% non-wetting oil. (c)–(d) Interpretation for the restricted diffusion oil signal in D - T_2 map and D - T_1 map of tight sandstone saturated with 25% wetting oil and 75% non-wetting water.

model is limited in tight sandstone, so we refer to the restricted diffusion model (Zielinski *et al* 2010, Minh *et al* 2012, 2015) and obtain D - T_2 correlation of wetting fluid in tight sandstone. D - T_2 correlation can be applied to identify wetting fluid in D - T_2 maps. We have further expanded the model to obtain D - T_1 correlation of wetting fluid in D - T_1 maps of tight sandstone. Under restricted diffusion, D - T_2 or D - T_1 correlation of wetting fluid is shown below:

$$D(T_{i,\{i=1,2\}}) = D_0 \left[1 - \gamma \frac{\alpha_i L_D + \left(\frac{L_D}{L_M}\right)^2}{\alpha_i L_D + \left(\frac{L_D}{L_M}\right)^2 + \gamma} \right], \quad (19)$$

where $\alpha_i = \frac{4}{9\sqrt{\pi}} \frac{1}{T_i} \frac{1}{\rho'_i}$, $\gamma = 1 - \frac{D_\infty}{D_0}$ and $L_D = \sqrt{D_0 T_D}$. For a CPMG pulse sequence, $T_D \approx \langle TE \rangle^2$, TE is echo spacing, $\langle \rangle$ denotes the average of the square root, L_M is the heterogeneity length scale of the medium, usually $L_M \gg L_D$. ρ'_i is the effective surface relaxivity, if there is only wetting fluid in the pore, $\rho'_i = \rho_i$. In the mixed-saturated case, $\rho'_i = \frac{S_{g,1}}{S_{g,1} + S_{1,2}} \rho_i$, where $S_{g,1}$ is the contact area between the wetting fluid and the grain surface, $S_{1,2}$ is the contact area between the wetting fluid and the non-wetting fluid.

The restricted diffusion model was applied to interpret D - T_2 and D - T_1 maps of tight sandstone saturated with 25% wetting fluid in section 4, as shown in figure 6. The blue dashed line represents the restricted water line, the green dashed line represents the restricted oil line, the white solid line represents

the bulk relaxation line of water, and the red solid line represents the bulk relaxation line of oil. The restricted wetting fluid line starts from the intersection point of its free diffusion line and bulk relaxation line, no matter whether in D - T_2 maps or in D - T_1 maps, such as point A in figure 6(a), point B in figure 6(b), point C in figure 6(c) and point D in figure 6(d). The wetting fluid apparent diffusion coefficient gradually decreases and tends to be constant with decreasing relaxation time (T_1 or T_2), and changing restricted fluid line tends to be the same regardless of the type of fluid. The wetting water signal sits on the restricted water line in figures 6(a) and (b), and the wetting oil signal sits on the restricted oil line in figures 6(c) and (d), which suggests that the wetting fluid signal can be accurately identified by the restricted diffusion model in D - T_2 maps and D - T_1 maps.

6. Conclusions

In this paper, magnetization decays of CPMG pulse sequences with different waiting times (TW) and echo spacings (TE) in tight sandstone were simulated, and D - T_2 - T_1 maps of tight sandstone under different fluid saturation were obtained by the multiple echo trains jointing inversion method. Various fluids saturations, wettability, data SNR and restricted diffusion effects on the NMR response of tight sandstone were thoroughly studied.

When the wettability of tight sandstone is determined, the surface relaxation rate of wetting fluid increases and the

phenomenon of restricted diffusion tends to be obvious with its saturation decreasing. Furthermore, the wetting fluid signal moves along the directions of short relaxation time and small diffusion coefficient in D - T_2 - T_1 maps. When the fluid saturation of tight sandstone is determined, if rock wettability is changed from water-wet to oil-wet, the wetting oil signal will shift to the short relaxation time in D - T_2 - T_1 maps due to the effect of surface relaxation. With decreasing SNR, the ability to identify water and oil signals based on NMR maps gradually decreases.

The wetting fluid signal will move away from the fluid free diffusion line in tight sandstone D - T_2 - T_1 maps because of the effect of restricted diffusion, but can be identified accurately by using the expanded restricted diffusion model.

Acknowledgments

This project was funded by the National Natural Science Foundation of China—China National Petroleum Corporation Petrochemical Engineering United Fund (U1262114) and the Specialized Research Fund for the Doctoral Program of Higher Education of China (Grant No. 20130007110012).

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